Question One: Choose the correct answer of the following transfer it in CAPITAL letter to table:

Q.1) All of the orbitals in a given subshell have the same value of the _____ quantum number.

A) principal B) angular momentum C) magnetic D) A and B E) B and C

Q.2) Consider the following electron configurations to answer the questions that follow:

(i) $1s^2 2s^2 2p^6 3s^1$
(ii) $1s^2 2s^2 2p^6 3s^2$
(iii) $1s^2 2s^2 2p^6 3s^2 3p^1$
(iv) $1s^2 2s^2 2p^6 3s^2 3p^4$
(v) $1s^2 2s^2 2p^6 3s^2 3p^5$

The electron configuration that belongs to the atom with the lowest second ionization energy is _____.

A) (i) B) (ii) C) (iii) D) (iv) E) (v)

Q.3) The formula weight of potassium dichromate ($K_2Cr_2O_7$) is _____ g/mol. *(Atomic masses (g/mol): K=39, Cr=52, O=16)*

A) 107 B) 255 C) 242 D) 294 E) 333

Q.4) How many molecules of carbon dioxide are in 308.0 grams of carbon dioxide CO_2? *(M.m CO_2 = 44g/mol)*

A) 3.499×10^{22} B) 2.107×10^{24} C) 4.214×10^{24} D) 9.274×10^{25} E) 4.081×10^{27}

Q.5) What are the concentrations (M) of Mg^{+2} and $C_2H_3O_2^-$ resulted by dissolving 0.600 mol $Mg(C_2H_3O_2)_2$ in water and diluting to 135 mL?

A) 0.444 and 0.889 B) 0.044 and 0.089 C) 0.889 and 0.444 D) 0.444 and 0.444 E) 4.440 and 8.890

Q.6) What volume (mL) of a concentrated solution of magnesium chloride (9.00 M) must be diluted to 350. mL to make a 2.75 M solution of magnesium chloride?

A) 2.75 mL B) 50.0 mL C) 45.0 mL D) 107.0 mL E) 350.0 mL

Q.7) Calculate the mass of $Cu(NO_3)_2$ required to prepare 350.00 mL of a 0.500 M solution? *Molar mass of $Cu(NO_3)_2$ = 188.0 g/mol.*

A) 7.83 g B) 14.1 g C) 23.5 g D) 32.9 g E) 47.0 g

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Q.8) According to first law of thermodynamics_____

A) mass and energy are mutually convertible
B) entropy of a system remains constant.
C) heat and work are mutually convertible
D) mass and light are mutually convertible
E) heat flows from hot substance to cold substance.

Q.9) A 0.85 g gummy sample is burned in a bomb calorimeter. The temperature started at 23.5 °C and leveled off at 27.3 °C. The heat capacity of the calorimeter to be 11.4 kJ/°C. Calculate the heat of combustion per gram of gummy sample.

A) 34.98 kJ/g B) 43.89 kJ/g C) 50.96 kJ/g D) 59.65 kJ/g E) 64.55 kJ/g

Q.10) Calculate ΔH for the reaction: $N_{2(g)} + 2O_{2(g)} \longrightarrow 2NO_{2(g)}$ ΔH =?
Using the following thermochemical equations:

$N_{2(g)} + O_{2(g)} \longrightarrow 2NO_{(g)}$ ΔH= +180 kJ
$2NO_{2(g)} \longrightarrow 2NO_{(g)} + O_{2(g)}$ ΔH= +112 kJ

A) −68.0 kJ B) +68.0 kJ C) −182.8 kJ D) +248.0 kJ E) −248.0 kJ

Q.11) A closed system is one in which_____

A) Mass does not cross boundaries, but energy may do so
B) Neither mass nor energy crosses the boundaries of the system
C) Both energy and mass cross the boundaries of the system
D) Mass crosses the boundary but not the energy
E) Thermodynamic reactions take place.

Q.12) Determine ΔG° at 25 °C for the reaction:

$$5CH_{4(g)} + 5O_{2(g)} \longrightarrow 5CO_{2(g)} + H_2O_{(g)}$$

If ΔG_f° (kJ/mol): $CH_{4(g)} = -50.72$; $H_2O_{(g)} = -228.6$; and $CO_{2(g)} = -394.4$

A) +101 kJ B) −347 kJ C) −856.1 kJ D) +1479 kJ E) −1947 kJ

Q.13) Calculate ΔS° for the reaction: $2Mg_{(s)} + O_{2(g)} \longrightarrow 2MgO_{(s)}$. If S° (J/mol): $Mg_{(s)} = 32.7$; $O_{2(g)} = 205.1$; and $MgO_{(s)} = 26.9$

A) +146.4 J/mol B) −146.4 J/mol C) +133.8 J/mol D) −216.7 J/mol E) +216.7 J/mol

Q.14) The oxidation number of nitrogen in the series, N_2O, N_2, NH_3, and NO_3^- respectively are:

A) 0, +2, +5, −3 B) +1, 0, −3, +5 C) 0, +1, +5, −3 D) +5, −3, 0, +1 E) −3, +5, 0, +1

Q.15) In which one of the following compounds is rhenium (Re) in the highest oxidation state?

A) $NaReO_4$ B) $ReClO$ C) Re_2O_3 D) $ReCl_5$ E) ReO_2

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Question Two: Write the final answer for the following questions:

Q.1) At maximum, a d-subshell can hold _____ electrons, and a p-subshell can hold _____ electrons f-subshell can hold _____ electrons.

Q.2) Atomic radius generally decreases as we move _____.

Q.3) As successive electrons are removed from an element, the size of atom _____.

Q.4) Silver nitrate $AgNO_3$ (M.m =169.87 g/mol) and aluminum chloride $AlCl_3$ (M.m= 133.34 g/mol) react with each other by exchanging anions: $3AgNO_{3\,(aq)} + AlCl_{3\,(aq)} \rightarrow Al(NO_3)_{3\,(aq)} + 3AgCl_{(s)}$ The mass in grams of AgCl (M.m= 143.32 g/mol) is produced when 5.22 g of $AgNO_3$ react with 8.73 g of $AlCl_3$ _____ g.

Q.5) When aqueous solutions of $AgNO_3$ and NaCl are mixed, AgCl precipitates. The balanced equation is: _____ → _____.

Q.6) How much heat is required to raise the temperature of an 850 gram block of aluminum from 22.8°C to 84.6°C? The specific heat of Al is 0.902 J/g°C. _____ kJ.

Q.7) Calculate $\Delta H°$ for the following reaction:

$$HBr_{(g)} \rightarrow H^+_{(aq)} + Br^-_{(aq)} \quad \Delta H° = ?$$

If $\Delta H_f°$, $HBr_{(g)} = -36.2$ kJ/mol; and $\Delta H_f°$, $Br^-_{(aq)} = -120.9$ kJ/mol. _____ kJ.

Q.8) ΔH and ΔE for the reaction: $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$ at constant temperature are related as: _____.

Q.9) Calculate the Gibbs free energy for a reaction at 273 Kelvin if the change in enthalpy is 19.07 kJ and the change in entropy is 90 J per Kelvin. _____ kJ.

Q.10) Identify the species being oxidized and reduced in the following reaction:

$3Hg^{2+} + Fe_{(s)} \rightarrow 3Hg_2 + 2Fe^{3+}$ _____.

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